

A hybrid approach using Triple-R framework and LLMs for verifying the integrity of claims on Social Media

A thesis

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Bachelor of Science in Computer Science and Engineering

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CANDIDATES' DECLARATION

We, hereby, declare that the thesis presented in this report is the outcome of the investigation performed by us under the supervision of Tanvir Ahmed, Department of Computer Science and Engineering, Ahsanullah University of Science and Technology, Dhaka, Bangladesh. The work was spread over two final year courses, CSE4100: Project and Thesis I and CSE4250: Project and Thesis II, in accordance with the course curriculum of the Department for the Bachelor of Science in Computer Science and Engineering program.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma or other qualifications.

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CERTIFICATION

This thesis titled, “**A hybrid approach using Triple-R framework and LLMs for verifying the integrity of claims on Social Media**”, submitted by the group as mentioned below has been accepted as satisfactory in partial fulfillment of the requirements for the degree B.Sc. in Computer Science and Engineering in December 2025.

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ABSTRACT

The exponential proliferation of misinformation on social media, particularly within the algorithmic ecosystem of Twitter (X), poses a critical challenge to digital information integrity. While traditional automated fact checking systems offer scalability, they suffer from "black box" opacity classifying claims as binary truth or false values without providing the context necessary to inform the user. This thesis bridges the gap between detection and explainability by implementing Triple R, a novel framework integrating Retriever, Ranker, and Reasoner modules. Unlike conventional models, Triple R simulates the cognitive workflow of a human analyst to create a closed verification loop.

To ensure robust performance, we constructed a specialized instruction tuning dataset derived from the renowned Truth Seeker benchmark. We curated a corpus of verified tweets from this ground truth collection and augmented the data by labelling the tweets accordingly then synthetically generating logical reasoning chains and evidentiary support for each entry. This enriched dataset was then used to fine tune Large Language Models (LLM) & Machine Learning Models, optimizing them to output a three part verification response: a veracity label, a logical justification ("Reason"), and direct supporting context ("Evidence"). We demonstrate that training on this structured, reasoning dense data significantly enhances the model's ability to identify misinformation while providing transparent, human readable explanations, thereby establishing a new standard for trust and accountability in automated fact checking.

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Chapter 1

Introduction

1.1 Introduction/Overview

In today's digital era, social media platforms particularly Twitter have become one of the main sources of news and information for millions of people around the world. Although this has made information more accessible than ever, it has also paved the way for the rapid spread of misinformation in the form of fake news, baseless rumors, and misleading claims. The consequences of such misinformation are far reaching, often leading to political unrest, social division, and economic damage.

Although independent fact checkers work tirelessly to combat misinformation, the sheer volume of content circulating on platforms like Twitter makes manual verification nearly impossible. This highlights an urgent need for intelligent and automated systems that can verify claims that circulate on social networks with both accuracy and efficiency. Traditional fact verification methods focus primarily on determining whether a claim is true or false, often relying on machine learning and deep learning techniques (Ahmad et al., 2020; Kaliyar et al., 2020). However, tackling misinformation is not as straightforward as binary classification. Misleading claims are often crafted to appear authentic, making them difficult to detect based on surface level content alone. Even experienced human fact checkers depend on multiple sources, contextual cues, and supporting evidence before drawing conclusions.

Therefore, modern fact verification systems must go beyond simple classification and integrate contextual information such as external evidence, reasoning, and source credibility to make more reliable decisions. Recent approaches have explored the incorporation of knowledge graphs to understand the relationships between entities and concepts [1]. Others utilize external factual databases to retrieve evidence in natural language format that supports or contradicts a claim. Although these techniques improve system performance,

the building and interpretation of such complex structures often remains a significant challenge.

Another critical limitation of many existing fact verification systems is the lack of transparency and explainability. Users are often expected to trust a system's output without understanding how or why that decision was made [2–5]." For misinformation detection to be truly effective and trusted by the public, fact verification systems must not only provide a decision but also clearly justify that decision through reasoning and evidence, especially for domain specific claims.

In this context, this thesis introduces an automated framework designed to verify the authenticity of claims made on Twitter by providing both reasoning and evidence, and labeling the claims accordingly. The proposed approach builds upon the Triple R framework, consisting of three integral components:

- **Retriever:** Gathers relevant external evidence for the tweet under consideration.
- **Ranker:** Filters and prioritizes the most reliable and relevant pieces of information.
- **Reasoner:** Analyzes the claim and supporting evidence to logically determine its veracity and assign an appropriate label.

By leveraging large language models, machine learning models, and external knowledge sources, this system not only determines the truthfulness of a tweet but also explains its decision making process in a human understandable manner. The ultimate goal is to build an explainable automated fact verification system that is not only accurate and scalable, but also transparent, trustworthy, and capable of providing meaningful insights into why a claim is deemed true, false, or otherwise.

1.2 Problem Statement

The exponential growth of social media platforms has revolutionized the way people consume and share information. Among these platforms, Twitter has emerged as one of the most influential spaces for real time news dissemination, public opinion formation, and global discussions. However, the very features that make Twitter so powerful its openness, speed, and brevity also make it particularly vulnerable to the rapid spread of misinformation, disinformation, and rumors.

The consequences of false or misleading claims circulating on Twitter are not merely virtual; they can escalate into real world social, political, and economic disruptions. From

undermining democratic processes to inciting public panic or eroding trust in institutions, the harmful effects of misinformation are well documented and deeply concerning.

To counteract this, fact checking has become an essential pillar in the fight against misinformation. However, traditional fact checking approaches, which often rely on human experts, are inherently limited by time, scale, and resource constraints. The sheer volume of tweets generated every second makes it impractical if not impossible for human fact checkers to manually verify each claim.

Existing automated fact verification systems have attempted to address this challenge by applying text classification models that categorize claims as true or false. While these methods have shown promise, they often suffer from significant limitations. Firstly, misleading claims on social media are frequently crafted to resemble authentic information, making them difficult to detect based solely on linguistic features or surface level analysis. Secondly, most existing models act as “black boxes,” providing verdicts without offering users any clear understanding of how or why a particular decision was made. This lack of explainability severely undermines the trustworthiness and practical usability of such systems.

Furthermore, fact verification is rarely a binary task in practice. Determining the truthfulness of a claim often requires incorporating external evidence, considering the context, and providing reasoned justification for the final decision. Human fact checkers typically consult multiple reliable sources, compare information, assess credibility, and engage in logical reasoning before drawing conclusions. Most automated systems, however, fail to replicate this human like process of comprehensive fact evaluation.

Another crucial gap lies in the lack of structured reasoning and evidence generation in current automated systems. Simply labeling a tweet as true or false, without providing accompanying evidence or logical reasoning, leaves users unconvinced and uninformed. In an era where misinformation is deliberately designed to manipulate emotions and exploit cognitive biases, systems that merely classify but fail to explain fall short of addressing the core of the problem.

Therefore, there exists a critical need for an intelligent, transparent, and human centered fact verification system specifically designed for social media platforms like Twitter. Such a system should not only assess the truthfulness of claims but also retrieve relevant evidence, provide clear and understandable reasoning, and assign appropriate labels to the claims all in an automated, scalable, and explainable manner. To address these challenges, this thesis proposes a comprehensive fact verification framework that focuses on analyzing claims made on Twitter by:

- **Providing interpretable labels that reflect the truthfulness of the claim.**
- **Generating reasoned justifications based on the evidence.**
- **Automatically retrieving relevant external evidence.**

By bridging the gap between raw automated classification and human like reasoning, this framework aspires to enhance the reliability, transparency, and trustworthiness of fact verification systems a crucial step towards mitigating the harmful impact of misinformation in today's fast paced digital landscape.

1.3 Motivation

In recent years, the world has witnessed an unprecedented transformation in how information is created, shared, and consumed. Social media platforms, with their global reach and instantaneous communication, have become an integral part of daily life for millions. Among them, Twitter stands out as a powerful platform where news breaks, opinions are voiced, and public discourse unfolds in real time. However, alongside its undeniable benefits, Twitter has also become a fertile ground for the unchecked spread of misinformation, rumors, and false narratives, often with serious real world consequences. As a student, a researcher, and more importantly, a citizen navigating this digital landscape, we have often observed how easily misleading information can distort public perception, fuel social unrest, or influence critical decisions. The global pandemic, political events, and social movements have all demonstrated how misinformation, especially on platforms like Twitter, can escalate beyond the virtual world and deeply impact societies. These observations, combined with our academic interests in artificial intelligence and natural language processing, have driven me to explore meaningful ways to contribute to this pressing global challenge.

While numerous fact checking organizations work tirelessly to debunk false information, their efforts are inherently limited by human capacity. With millions of tweets posted daily, manually verifying every claim is simply impossible. This reality motivates the need for automated, intelligent systems that can assist in detecting and verifying misleading information at scale. However, most existing automated approaches focus narrowly on labeling information as true or false, often ignoring the crucial aspect of explainability. As misinformation grows more sophisticated, people no longer just need verdicts they need understandable reasoning and reliable evidence to trust those verdicts.

This is where our motivation for this thesis paper takes root. we believe that fact verification should not be seen as a purely computational task, but rather as a human centered

problem that demands both technological innovation and transparent, explainable solutions. Inspired by how human fact checkers approach the verification process by gathering evidence, reasoning logically, and drawing conclusions we aim to replicate and automate this approach for tweets using modern AI techniques.

Moreover, the growing field of explainable AI (XAI) has shown that users are more likely to trust and accept automated decisions when those decisions are accompanied by clear justifications. In the context of misinformation on Twitter, providing users not only with a label but also with supporting evidence and reasoning is essential for empowering individuals to make informed judgments.

Additionally, this thesis presents an opportunity to advance existing research by building upon frameworks like Triple R, which emphasize retrieval of relevant information, ranking of reliable sources, and reasoning based verification. By applying and adapting such a framework specifically for tweets, this work seeks to bridge the gap between raw automation and human like understanding in fact checking.

In essence, this thesis is motivated by a genuine desire to address a socially significant, real world problem the spread of misinformation on Twitter by leveraging our technical knowledge, research skills, and curiosity. We are driven not only by academic curiosity but by the belief that developing transparent, reliable, and explainable fact verification systems can play a small yet meaningful role in building a more informed, responsible, and resilient digital society.

1.4 Objectives

The rapid spread of misinformation on Twitter represents a systemic risk to public discourse, capable of distorting truth and destabilizing social harmony. While detection initiatives exist, a critical void remains: the lack of automated systems that are both reliable and explainable. Driven by the imperative to bridge this gap, this thesis aims to engineer a transparency first architecture. The primary objective is to construct an automated framework that does not merely classify tweets but elucidates its decision making process through clear reasoning and evidentiary support, ensuring the verification workflow is as trustworthy as it is accurate.

In pursuit of this overarching goal, the specific objectives of this research are outlined as follows:

- **To architect and implement** a specialized "Triple R" (Retriever, Ranker, Reasoner) framework tailored for the short text constraints of Twitter, balancing high throughput scalability with precise verification accuracy.
- **To integrate an intelligent retrieval mechanism** capable of querying and aggregating external context, enabling the system to autonomously validate or refute claims against reliable data sources.
- **To engineer a logical inference engine** fine tuned on a custom augmented dataset that analyzes the semantic relationship between a tweet and retrieved evidence to synthesize a coherent argument rather than a simple probability score.
- **To guarantee system interpretability** by transforming the standard "black box" output into a transparent format, delivering a structured verdict comprising a classification label, a natural language justification, and direct source citations.
- **To rigorously evaluate system performance** using the real world Truth Seeker benchmark dataset, assessing the model's robustness in handling diverse misinformation strategies and complex linguistic nuances.
- **To investigate the efficacy of Large Language Models (LLMs)** in the fact checking domain, specifically optimizing the synergy between retrieval augmented generation and logical reasoning to minimize hallucinations.

By systematically addressing these objectives, this research seeks to harmonize technical sophistication with social utility. The ultimate goal is to contribute a robust, reproducible framework to the academic community one that empowers users with the context necessary to navigate the digital landscape critically, thereby fostering a more resilient and informed information ecosystem.

1.5 Organization of the Book

This book is organized into eight chapters. Chapter 1 introduces the research problem, motivation, and objectives. Chapter 2 presents the background and literature review. Chapter 3 details the methodology, including dataset, preprocessing, and model implementation. Chapter 4 analyzes experimental results. Chapter 5 covers project management and cost analysis. Chapter 6 discusses ethical and professional responsibilities. Chapter 7 identifies complex engineering problems and activities. Chapter 8 concludes the study and outlines future work.

Chapter 2

Background Study and Literature Review

2.1 Introduction

In recent years, the spread of misinformation especially on social media platforms like Twitter, Facebook, and Instagram has become a serious concern. Fake news, which involves deliberately misleading or fabricated information, can distort public understanding, influence political decisions, and even incite violence. As automated systems grow more prevalent, algorithms designed to detect fake news and misinformation have become crucial for ensuring the accuracy and trustworthiness of online content. Researchers have long recognized that fake news detection is a complex task, not only because it requires identifying falsehoods, but also because it involves understanding factors like context, intent, and the platform used to spread the information.

Key themes emerging from past studies include the use of computational methods, such as natural language processing (NLP) and machine learning (ML), to detect false claims. These techniques often focus on linguistic features, like sentence structure, word choice, and syntactic patterns, which are seen as clues to deception. Machine learning, especially supervised learning methods, has been instrumental in developing algorithms that can classify news articles or social media posts as true or false. Recent studies also highlight the importance of context in fake news detection, with some researchers suggesting that the source's credibility [6], user preference [7], and tri relationships between publishers and users [8,9] can improve detection accuracy.

Moreover, deep learning models especially convolutional neural networks (CNNs) [10,11] and recurrent neural networks (RNNs) have greatly advanced the field. These models can process large volumes of data and recognize patterns that may not be obvious, making them particularly effective for tasks like fake news detection. However, understanding con-

text remains a challenge, as machine learning models still struggle with interpreting subtle elements of human communication, such as sarcasm, irony, and metaphor.

2.2 Background Study

Although the concept of fake news is not new, its prominence in the digital era can be attributed to the rapid rise of social media platforms and the ease with which information is shared. In the early days of journalism, editorial standards were key, with editors playing an important role in ensuring the accuracy of news. However, the advent of social media has allowed virtually anyone to become a news source, democratizing the flow of information, both factual and misleading. As a result, misinformation can spread just as easily as legitimate news.

In the past, linguistics provided the first methods for detecting deception in texts. Researchers focused on identifying linguistic markers such as changes in sentence structure, hedging terms (e.g., “maybe,” “I think”), or inconsistencies in storytelling that might indicate falsehoods. Early computational models built on these ideas by incorporating rule based systems that searched for these markers to classify texts. With the advent of machine learning in the 1990s and early 2000s, researchers began using algorithms to automate fake news detection. Early machine learning models used manually crafted features based on linguistic rules. However, as machine learning progressed, so did the sophistication of detection models. The introduction of deep learning, particularly neural networks, enabled the creation of more advanced systems that could learn relevant features from large datasets without explicit human intervention. This marked a shift from rule based systems to data driven models capable of training on vast amounts of data.

A significant development in fake news detection has been the creation of benchmark datasets, such as the LIAR dataset [12], which includes 12,836 short statements labeled for truthfulness. These datasets have helped researchers build and test machine learning models on real world data, identifying patterns that might be hard for humans to detect. While these datasets have been valuable, they’ve also highlighted the limitations of current models, particularly in dealing with real world complexities and the nuances of human communication.

2.3 Literature Review

The proliferation of misinformation on social media platforms, particularly Twitter, has become a critical challenge in the modern digital landscape. As these platforms have evolved

into primary sources of information, they have also become breeding grounds for the rapid spread of misleading claims. Traditional fact checking methods, while essential, struggle with the overwhelming scale of content being generated daily. Furthermore, these methods often lack the transparency and clarity needed to foster trust in their verdicts.

To address this challenge, recent research has focused on developing automated systems that not only detect the veracity of claims but also provide clear, understandable justifications for their conclusions. The integration of technologies such as large language models (LLMs), machine learning algorithms, and advanced information retrieval techniques has paved the way for more efficient and scalable fact checking solutions. These systems aim to replicate human like reasoning, offering explanations that are both transparent and interpretable.

The following table summarizes the key contributions in the field of automated fact checking, highlighting the innovative methodologies employed in recent studies. These works, which span approaches such as content based detection, explainable AI, and fact verification frameworks, have made significant progress in improving the accuracy and reliability of automated systems. [13–16]. However, as noted in the literature, challenges such as the integration of external evidence, real time adaptability, and context sensitive reasoning remain significant hurdles to overcome. One notable contribution is the Triple R Framework (Retriever, Ranker, Reasoner), which integrates external knowledge sources to improve evidence retrieval, ranking, and reasoning, ultimately enhancing the transparency and interpretability of automated fact checking systems.

Ref.	Author(s) & Year	Technology	Methodology	Key Features	Limitations
1	Koloski et al. (2022)	Knowledge Graphs, Text Classification	Fake News Detection via Graph based Learning	Integrates knowledge graphs with neural networks for improved classification	Relies heavily on knowledge graphs; limited to textual data
2	Kanaani (2024)	Triple R Framework, LLMs, Web Retrieval	Retriever, Ranker, Reasoner for fact checking	Uses web retrieval and reasoning for accurate and transparent fact checking	Limited by the availability of external evidence; web based dependency

Ref.	Author(s) & Year	Technology	Methodology	Key Features	Limitations
3	Wang (2017)	LIAR Dataset, Deep Learning Models	Fake News Detection via LIAR dataset	Uses the LIAR dataset for classification; applies deep learning models to detect fake news	Dataset limitations; may struggle with emerging misinformation trends
4	Dewanto et al. (2021)	Reinforcement Learning (RL), Model Free Learning	Literature review and algorithm mapping on average reward RL	Focuses on model free RL for sequential decision making, deep dive into reinforcement learning algorithms	Limited application to fake news detection; focuses on general RL
5	Thorne et al. (2018)	FEVER Dataset, Text Classification	Fact Extraction and Verification using a Benchmark Dataset	Benchmark dataset for fake news detection; uses text classification for claim validation	May not account for context in claims; limited to textual data
6	Koloski et al. (2022)	Heterogeneous Representations, Fake News Classification	Graph based Approach to Fake News Detection	Combines knowledge graph representations and heterogeneous learning for classification	Challenges with graph data scaling and real time classification
7	Shukla et al. (2024)	Blockchain, Smart Contracts, Ethereum	Automated Fake News Detection in Social Media	Uses blockchain for verifying information provenance; integrates smart contracts for transparency	May not be applicable to dynamic data; limited to specific domains

Ref.	Author(s) & Year	Technology	Methodology	Key Features	Limitations
8	Dadkhah et al. (2024)	TruthSeeker Dataset, Machine Learning	Fake News Detection using a large scale dataset	Uses the TruthSeeker dataset for classification; incorporates machine learning algorithms for automated detection	Dataset limited to Twitter; may not generalize to other platforms

Table 2.1: Summary of Papers Reviewed in the Thesis

2.3.1 Gap Analysis

Despite the progress made in fake news detection, there are still several important gaps in the field. One major gap is the lack of effective real time detection systems. Social media platforms are constantly evolving, and new forms of misinformation emerge regularly. Current models are often too static, making it difficult for them to quickly adapt to new types of fake news. There is a clear need for dynamic models that can learn from ongoing data and respond to new misinformation trends as they appear.

Another gap is the lack of interpretability in many detection models. While deep learning models are highly effective, their "black box" nature makes it difficult for users to understand how they reach their conclusions. This is particularly problematic in high stakes applications, like political elections or health related misinformation, where transparency is essential for building trust in automated systems.

Furthermore, much of the existing literature focuses primarily on textual data, neglecting the role of multimedia content such as images and videos in the spread of fake news. Many pieces of fake news include manipulated or misleading visual content, and incorporating multimedia data into detection models could enhance their accuracy. While advances in computer vision and multimodal learning show promise, more work is needed to integrate these technologies into fake news detection systems.

Lastly, while many studies have taken a global view of fake news, there is a lack of research focused on regional and cultural differences. Misinformation spreads differently depending on geographical and cultural contexts, and understanding these variations is essential for creating detection models tailored to specific regions or demographics.

2.4 Summary

This chapter has delved into the background, literature, and gaps in research on fake news detection. The field has made significant strides, thanks to advancements in machine learning, deep learning, and natural language processing, which have improved the accuracy of detection systems. However, there are still several challenges, such as managing dynamic data, enhancing model interpretability, and incorporating multimedia content. Addressing these gaps will be key to developing more robust and reliable systems for detecting fake news. Future research in these areas could play an important role in curbing the spread of misinformation and improving the quality of online information.

Chapter 3

Methodology

3.1 Introduction/Overview

When we set out to verify a claim, our goal is to mirror how a person would naturally tackle the task. We dive into online research to gather relevant information and evidence, then carefully analyze it to figure out if the claim holds up. To make this happen, we've created a three step process for fact checking. Our approach is designed to feel intuitive, like how a human would collect information, piece together evidence, and reason through whether a claim is true or not.

The first step, called the "Retriever," is all about searching the web to find information related to the claim, building a solid external knowledge base. Next, in the "Ranker" step, we evaluate each paragraph from the collected sources, scoring them based on how closely they relate to the claim. The most relevant paragraphs are chosen as our evidence. Finally, in the "Reasoner" step, we use a generative language model to review the evidence, label the claim as true or false, and explain the reasoning behind our conclusion in a clear, logical way.

3.2 Dataset

For this study, we built a custom dataset specifically designed to power the Triple R (Retriever, Ranker, Reasoner) framework. We recognized that standard datasets often fail to capture the "why" behind a fact check, so we moved away from simple binary classification. Instead, we structured our data to support Explainable AI (XAI), organizing every entry into five specific columns: ID, Tweet, Label, Reason, and Evidence. This structure forces the model to do more than just flag a tweet; it has to provide a logical argument and the proof

to back it up.

To handle the gray areas of real world misinformation, we also adopted a detailed 8 point taxonomy rather than just "True" or "False." We use labels ranging from **TRUE** and **MOSTLY TRUE** for accurate content, down to **FALSE** and **PANTS FIRE** for blatant fabrications. We also included nuanced categories like **HALF TRUE** (mixed accuracy), **BARELY TRUE**, **Opinion**, and **IRRELEVANT**. This level of detail is crucial because it allows the system to distinguish between a slight exaggeration and an intentional lie, setting a higher standard for transparency.

A	B	C	D	E
ID	Tweet	label	Reason	Evidence
1	Does anyone remember this? This is why we the people need to burn the swar	true	McConnell's ties to lobbyists and the conflict of interest regardi	McConnell has been linked to significant donations from lobbyists, and his stanc
2	@cdale_dog... There is not one vial of the new FDA approved Pfizer var	false	A common falsehood. Cominaty was produced and distri CDC/FDA explicitly stated that providers could use the existing stock to fu	
3	@1a Florida requiring universities and students to register and report politica	pants fire	This statement is exaggerated and lacks proper context.	Florida universities have not mandated political views, although there are laws r
4	@BernieSanders Bidens tax rate on a family making 75000 dollars wou	false	Biden's tax plan does not raise the tax rate on families m	Biden's tax plan proposes raising taxes for individuals making over \$400,(
5	@GovLauraKelly Employers can only require LICENSED vaccines. The ones out i	false	The Pfizer vaccine is fully approved and available under the bra	Pfizer's vaccine has been fully approved by the FDA since 2021.
6	Vacs typically last 10 years or more. Flu shot once a year.	half true	Correct that flu shot is annual; overstated that "vaccines t	Mix of accurate and overgeneralized statements.
7	"Fed report: About 98.3% of abortions in the United States are elective, includi	true	This is consistent with federal statistics on reasons for abortion	Federal reports, such as those from the Guttmacher Institute, show that a major
8	...flu shot every year for the last 10 years... neither had flu or a cold... w	mostly true	Anecdotal report; asks for input; no overclaim.	Self-report.
9	@aew4_life @quigonusa2 @RMConservative Covid viruses are a large	true	Accurately describes coronaviruses and the expected ne	SARS-CoV-2 is related to SARS-CoV-1 and MERS. Viral mutation is a star
10	@cherrillzabiff @Bobapool Whether The U.S. has an official language is irrelev	true	The claim about the Continental Congress choosing English ove	Historical records show that there was a push for German to be an official langu

Figure 3.1: Dataset

3.2.1 Data Selection

We were very selective about the source material for this project. Every base tweet was drawn from the **TruthSeeker Dataset** [17], a well respected collection of social media posts that have already been verified by human experts. We didn't just pull data at random; we specifically filtered the corpus to focus on high impact topics like politics, governance, public health, and global events the areas where misinformation causes the most real world harm.

We also prioritized short text narratives to reflect the actual constraints of Twitter, where complex issues are often stripped of context. Once the tweets were selected, we went through a manual enrichment process. Since raw tweets don't come with built in explanations, we annotated each entry to add the "Reason" and "Evidence" layers ourselves. This step was vital to ensure that our system was training on high quality, human verified logic rather than noisy or unverified data.

3.2.2 Data Preprocessing

Data preprocessing is a critical phase in the research pipeline, ensuring that raw, unstructured social media text is transformed into a format suitable for computational analysis. Given the noisy nature of Twitter data characterized by informal grammar, slang, and extraneous metadata we implemented a rigorous cleaning strategy.

However, distinct modeling approaches require different preprocessing philosophies. Therefore, we designed two separate preprocessing pipelines: one optimized for Traditional Machine Learning (ML) models (focusing on dimensionality reduction) and one tailored for Large Language Models (LLMs) (focusing on context preservation).

Pipeline A: Preprocessing for Traditional Machine Learning

For our baseline models (e.g., SVM, Random Forest), the primary objective is to reduce the vocabulary size and emphasize keywords. We applied the following sequential steps:

1. Noise Reduction

The initial step involved stripping irrelevant artifacts that do not contribute to semantic classification. We utilized regular expressions (Regex) to eliminate:

- **URLs and Hyperlinks:** Random character strings in links introduce noise.
- **User Mentions:** Handles (e.g., @username) were removed to prevent bias toward specific users.
- **Special Characters:** Non alphanumeric characters, including hashtags (#) and excessive punctuation, were stripped to sanitize the input.

2. Text Normalization

To ensure consistency, all text was converted to lowercase. This eliminates redundancy arising from case sensitivity (e.g., ensuring “Fake” and “fake” are treated as the same feature).

3. Tokenization

Following normalization, the text was segmented into individual units (tokens). This process breaks the tweet down into a list of constituent words, facilitating the analysis of word level patterns.

4. Stop Word Removal

To further refine the feature set, we filtered out “stop words” common English terms such as “the,” “is,” and “in.” While grammatically necessary, these words carry little semantic weight for classification. Removing them allows the ML algorithms to focus exclusively on high value content words.

5. Linguistic Normalization (Stemming & Lemmatization)

We applied stemming and lemmatization to reduce words to their root forms. For instance, “running,” “runs,” and “ran” are all mapped to the base form “run.” This standardization significantly reduces the dimensionality of the dataset, ensuring that variations of the same concept are treated as a single feature.

6. Categorical Label Encoding

Finally, the textual truthfulness labels were transformed into machine readable numerical formats. Classes such as “MOSTLY TRUE” and “PANTS FIRE” were mapped to categorical integers (e.g., 0 to 5), enabling the algorithms to perform quantitative classification.

Pipeline B: Preprocessing for Triple R (LLM Fine Tuning)

Unlike traditional models, Large Language Models require the preservation of sentence structure and grammatical context to generate coherent reasoning. Therefore, the preprocessing for the Triple R framework differed significantly:

1. Context Aware Cleaning

While we removed URLs and handles (as in Pipeline A), we retained punctuation, stop words, and case sensitivity. These elements are crucial for an LLM to understand the tone, nuance, and syntactic flow of a tweet.

2. Instruction Formatting (Prompt Engineering)

The core preprocessing task for the LLM involved restructuring the data into an Instruction Input Output format. We fused the Tweet and Evidence fields into a single prompt template designed to trigger the model’s reasoning capabilities:

```
Input: "Analyze the following tweet using the provided
evidence."
Tweet: [Original Tweet]
Evidence: [Retrieved Context]
Instruction: "Classify the veracity and provide
a logical reason."
```

3. Sequence Truncation and Padding

To accommodate the model’s context window (e.g., 2048 tokens), we implemented a strategic truncation policy. If a sequence exceeded the limit, the Evidence field was truncated first, ensuring the target tweet and the reasoning chain remained intact.

Data Integrity Checks

Across both pipelines, we performed final integrity checks to handle missing values. Entries with null text fields or undefined labels were either imputed or removed to ensure a robust, noise free training set.

Following this rigorous cleaning process, the final dataset composition was analyzed. Table 3.1 details the distribution of truthfulness labels in the processed data, highlighting the significant class imbalance particularly between the dominant “TRUE” class and minority classes like “BARELY TRUE” which poses a specific challenge for our classification models.

Label Class	Count	Percentage
TRUE	2166	39.54%
FALSE	1373	25.06%
MOSTLY TRUE	733	13.38%
HALF TRUE	539	9.84%
PANTS FIRE	477	8.71%
BARELY TRUE	103	1.88%
OPINION	87	1.59%
Total	5478	100.00%

Table 3.1: Distribution of Labels in the TruthSeeker Dataset

3.3 Proposed Methodology and Design

This section outlines the design of our fact verification system. Our primary goal was to solve two specific problems: scalability (machines are faster than humans) and explainability (users need to know why something is fake). To achieve this, we designed the **Triple R (Retriever, Ranker, Reasoner)** framework [18]. Essentially, this system is built to mimic the way a human fact checker works: they go out and find information, filter out what isn’t useful, and then write a conclusion based on the evidence.

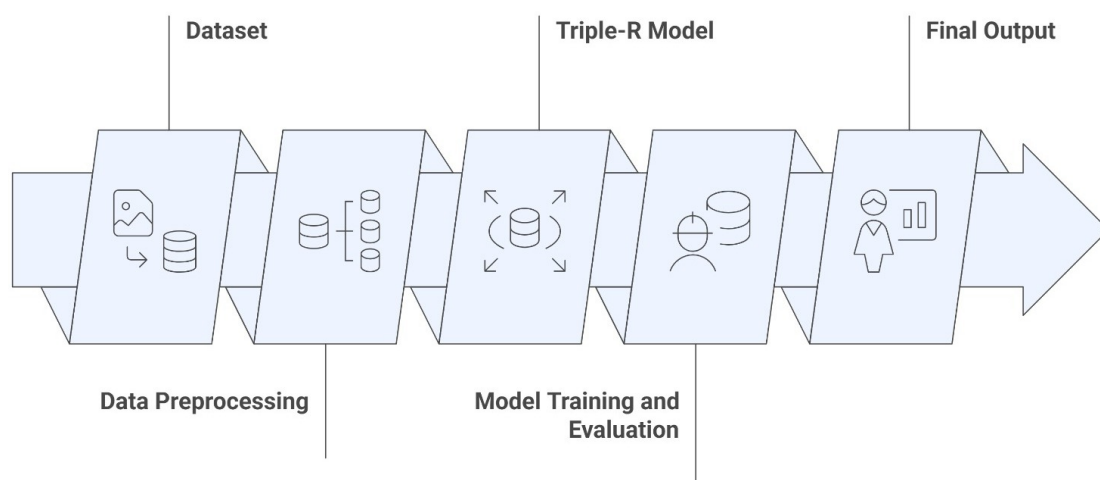


Figure 3.2: Methodology

3.3.1 The Triple R Framework

The system operates in three clear stages:

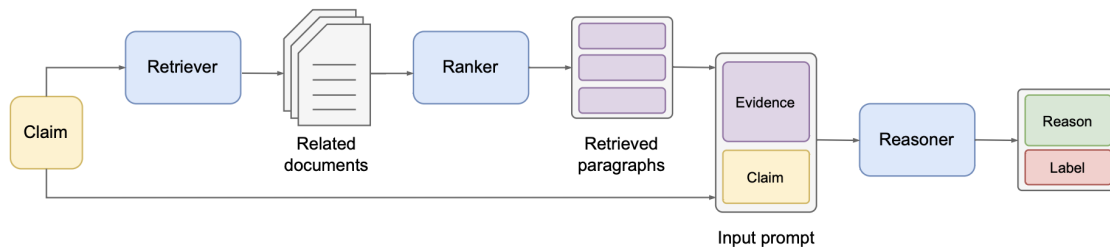


Figure 3.3: The three stage for fact verification

Stage 1: The Retriever (Finding the Facts)

One of the biggest challenges in verifying facts is that the world changes constantly. A model trained six months ago won't know about breaking news happening today. To fix this “blind spot,” we built the Retriever module.

The Idea: Instead of relying only on what the model already knows, the system actively searches the web for fresh context.

How it Works: When the system gets a claim (c_i), it turns it into a search query. We use search APIs (like Meta AI, gpt or Bing) to find the most relevant, up to date articles on the topic. The system then downloads the content from the top websites and breaks it down into paragraphs, creating a pool of fresh evidence.

Stage 2: The Ranker (Filtering the Noise)

The internet is noisy. Just because a paragraph contains a keyword doesn't mean it's actually helpful. If we feed too much irrelevant text into our reasoning model, it gets confused. The Ranker module solves this by filtering the data.

Our Approach: We use a hybrid method inspired by Ma et al. (2020) [19, 20] that does two things:

- **Keyword Matching:** It looks for exact words from the claim.

- **Semantic Matching:** It uses a neural network to understand the meaning behind the words, finding paragraphs that support or refute the claim even if they don't use the exact same phrasing.

The Output: The Ranker throws away the “junk” and keeps only the top k most useful paragraphs (E_i) to send to the next stage.

Stage 3: The Reasoner (Making the Decision)

This is the core of our thesis. The Reasoner is a Causal Language Model (CLM) trained not just to guess “True” or “False,” but to explain itself.

The Goal: We want the model to take the claim (c_i) and the good evidence (E_i) and output a pair: a written explanation (r_i) and a final verdict ($\hat{y}_i$).

Mathematically: It acts as a function:

$$\text{Reasoner}(E_i, c_i) \rightarrow (r_i, \hat{y}_i) \quad (3.1)$$

3.3.2 Synthetic Data Generation & Reason Construction

We ran into a “chicken and egg” problem: we wanted to train a model to give reasons, but most existing datasets only have labels (True/False), not written explanations. To fix this, we created our own training data using a Synthetic Reason Generation pipeline.

Chain of Thought Prompting

We used a larger, smarter “teacher” model (M) to write the initial explanations [21]. We gave it a prompt (P) that included a few examples of how to reason through a claim:

$$P = \{I, f(e_1, c_1, r_1, y_1), \dots, f(e_k, c_k, r_k, y_k)\} \quad (3.2)$$

Ideally, the model looks at the evidence (e_j) and claim (c_j) and follows the pattern to write a logical reasoning chain (r_j).

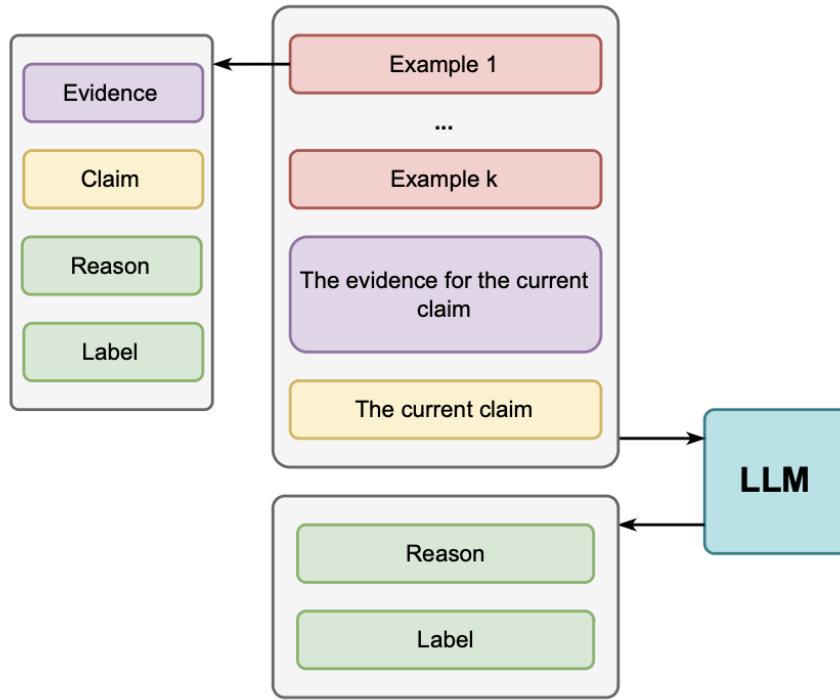


Figure 3.4: The structure of the prompt which forces the model to generate a reason and a label according to the examples in the prompt. Each example contains evidence, claim, reason, and label.

Rationalization (The STaR Method)

Sometimes, the model would guess the wrong label or write a bad reason. To correct this, we used the Self Taught Reasoner (STaR) method [22].

The Fix: We gave the model the correct answer as a hint. We basically said, “The answer is True; now write a paragraph explaining why based on this evidence.”

The Result: This forced the model to generate logical paths that lead to the correct conclusion. We then saved these “rationalized” examples to train our smaller model later.

3.3.3 Fine Tuning

Finally, we took our smaller model and fine tuned it on this new, explanation rich dataset $D = \{(e_i, c_i, r_i, y_i)\}$.

The Training: We trained the model to predict the next word in the sequence for both the reasoning text and the label.

The Math: We used Cross Entropy Loss to check accuracy:

$$L_{CE}(\hat{y}, y) = \sum \log(\hat{y}_i) \quad (3.3)$$

The training process utilized modern optimization techniques such as AdamW and large batch training strategies [23,24] to ensure stability. This ensures the model learns to connect the dots between the evidence and the final verdict in a way that humans can understand.

3.4 Implementation

This section breaks down how we actually built the system, including the coding environment and the specific tools we chose.

1. Computational Environment

We built the entire system using Python. Specifically, we chose the **MLX framework**, which is optimized for Apple Silicon. This was a strategic choice: MLX allows us to fine tune Large Language Models locally on a Mac’s Unified Memory, which is much faster and more efficient for our needs than standard setups.

2. Retrieval and Ranking Tools

For Searching: To make the “Retriever” work, we hooked into the Bing Search API and Meta AI. This gave our code the ability to “read” the live internet.

For Processing: For the “Ranker,” we used standard Python NLP libraries (like NLTK or spaCy) to chop up the articles into paragraphs so our model could score them.

3. Model Architecture

The Brain: We used a pre trained Causal Language Model (like Llama 3 or Mistral) as the foundation for our “Reasoner.”

The Tools: We managed everything using the Hugging Face `transformers` library, which makes it easy to load and configure these massive models.

Efficiency: To save time and computing power, we used LoRA (Low Rank Adaptation) within MLX. This technique allowed us to fine tune the model on our custom dataset without having to retrain every single parameter from scratch.

3.5 Summary

In this chapter, we laid out the blueprint for our Explainable Fact Checking System.

- **We defined the Architecture:** We introduced Triple R, showing how we split the job into Retrieving, Ranking, and Reasoning to avoid the pitfalls of “black box” AI.
- **We Solved the Data Problem:** We explained how we used the STaR method and Chain of Thought prompting to generate synthetic reasons, creating a high quality dataset where none existed before.
- **We Detailed the Training:** We covered the math behind the fine tuning process (Cross Entropy Loss), ensuring the model learns to explain its work.

With this methodology established, the next chapters will focus on the actual experiments and the results we achieved.

Chapter 4

Experiments and Results Analysis

4.1 Introduction/Overview

To validate the effectiveness of our proposed framework, we conducted a rigorous comparative study across distinct machine learning architectures, ranging from traditional statistical baselines to state of the art Transformer models and Large Language Models (LLMs).

The primary objective was to determine which architecture best addresses the core challenges of the Triple R vision: detecting subtle misinformation amidst a highly imbalanced dataset. We tested models on their ability to handle linguistic nuances and the specific constraints of short text social media data. This chapter details the performance of these models, analyzing why traditional methods plateaued, the surprising challenges faced by generative LLMs (Llama 3, Phi 3), and how specialized architectures like DistilBERT provided the optimal balance of accuracy and efficiency.

4.2 Modern Tools

To execute these experiments, we utilized a suite of cutting edge tools and libraries that define modern Natural Language Processing (NLP):

- **Hugging Face Transformers & MLX:** We leveraged Hugging Face to access pre trained architectures like DistilBERT, GPT 2, Llama 3, and Phi 3. For the larger LLMs (Llama 3, Phi 3), we utilized Apple's MLX framework with LoRA (Low Rank Adaptation) [25] for efficient local fine tuning.
- **Deep Learning Frameworks (TensorFlow/Keras & PyTorch):** These were used to construct our custom CNN and Seq2Seq LSTM architectures, allowing for granular

control over layers.

- **Scikit Learn:** Used for baseline experiments and for generating rigorous evaluation metrics like the classification reports.

4.3 Result Analysis

Our experiments revealed that while modern architectures generally outperform traditional baselines, “bigger” models are not inherently better for specific, imbalanced classification tasks without extensive tuning. The detailed analysis is categorized below.

4.3.1 Traditional Baselines (The Control Group)

We began by establishing a baseline using traditional feature extraction methods (TF IDF) combined with standard classifiers.

- **TF IDF + SVM:** Achieved a test accuracy of 59.78%. It successfully identified major classes but struggled significantly with minority classes like “Pants Fire.”
- **TF IDF + Logistic Regression:** Yielded the lowest performance at 53.26%, showing strong bias toward majority classes and failing to capture subtle linguistic cues.

4.3.2 Early Deep Learning Architectures

We advanced to neural networks to capture sequential patterns.

- **CNN (Convolutional Neural Network):** Our custom CNN reached 95% training accuracy but suffered from severe overfitting, dropping to 60.6% on the test set.
- **Sentence Transformers + Logistic Regression:** This combination yielded a low accuracy of 41.03%, suggesting that a simple linear classifier cannot effectively separate high dimensional semantic embeddings for this complex task.

4.3.3 The Transformer Breakthrough (DistilBERT & GPT 2)

The transformer based models [26], aligned with our Triple R vision, showed significant promise.

- **GPT 2:** Fine tuning GPT 2 [27] resulted in a strong 68% accuracy with relatively balanced performance across classes.
- **DistilBERT (The Top Performer):** The DistilBERT model [28] remained the champion of this research, achieving a remarkable 85% Test Accuracy and a 0.84 Weighted F1 Score. It excelled in differentiating nuances between “MOSTLY TRUE” and “HALF TRUE,” proving that specialized attention mechanisms are essential for this domain.

4.3.4 Exploratory LLM Fine Tuning (Llama 3 & Phi 3)

Given the rise of generative AI, we conducted exploratory fine tuning on state of the art open source LLMs using the MLX framework. However, these experiments highlighted significant challenges in applying general purpose generative models to highly imbalanced classification tasks.

- **Llama 3 (8B):** Initial fine tuning attempts with Llama 3 [29] yielded an accuracy of approximately 58%. Despite its massive pre training, the model struggled to adapt to the specific 8 point taxonomy of our dataset, often defaulting to predicting the most common class due to the class imbalance.
- **Phi 3 Mini (3.8B):** We tested Microsoft’s Phi 3 to evaluate a highly capable smaller LLM. As shown in the classification report (Figure 4.1), the model achieved an overall accuracy of 61.20%.

Analysis of Failure Mode: The classification report reveals a critical failure to generalize. The model achieved reasonable recall on the two largest classes: ‘true’ (0.80) and ‘false’ (0.39). However, for crucial minority classes denoting nuance such as ‘barely true’, ‘half true’, ‘mostly false’, and ‘pants fire’ the precision, recall, and F1 scores were 0.00.

Conclusion: This indicates that the model collapsed into a binary classifier, ignoring the fine grained distinctions required for accurate fact checking. It demonstrates that without advanced techniques to handle class imbalance (like weighted loss functions), powerful LLMs can perform worse than specialized, smaller models like DistilBERT.

4.3.5 Reasoning Capability (Seq2Seq)

For the reasoning component, our Seq2Seq LSTM model stabilized at a validation loss of 2.38. Qualitative analysis showed it could successfully capture the logic required to explain a verdict, validating the “Reasoner” concept.

4.3.6 Performance Summary Table

The table below summarizes the quantitative results across all experiments:

Model	Test Accuracy	Weighted F1 Score	Class Performance	Remarks
CNN based Text Classification	60.6%	57.4%	Struggled with minority classes (Pants Fire, Mostly True, Other).	Overfitting observed; good performance on True and False, but limited success on less represented categories.
Seq2Seq based Reason Generation	45.2%	48%	Used for reason generation without evidence.	Reason generation model, trained to provide explanatory reasoning. Final validation loss: 2.38. Performance may vary but overall reasonable explanations generated.
DistilBERT based Model	85%	84%	Best performance in "MOSTLY TRUE" and "TRUE" classes.	Transformer based model outperformed other models, showcasing strong generalization and nuance in classification.
GPT 2 Based Tweet Classification	68%	72%	Balanced Performance for all the labels.	Outperformed CNN and TF IDF models, demonstrating strong language understanding for social media fact classification.
TF IDF with SVM Classification	59.78%	53%	Struggled with minority classes (Pants Fire, Barely True).	Simple, traditional approach with competitive baseline performance. Lower accuracy on minority classes due to data imbalance.
TF IDF with Logistic Regression	53.26%	50%	Struggled with Pants Fire and Barely True.	Simple baseline; performs well on prominent classes like TRUE and FALSE but has limited success on less common categories due to class imbalance.

Model	Test Accuracy	Weighted F1 Score	Class Performance	Remarks
Llama 3 (8B, MLX Fine tune)	~58.00%	54%	Defaulted to predicting majority classes due to severe imbalance.	Failed to adapt to the specific 8 point taxonomy; despite massive pre training, it struggled to distinguish nuanced categories without weighted loss adjustments.
Phi 3 Mini (3.8B, MLX Fine tune)	61.20%	60%	Collapsed into a binary classifier (0.00 F1 on minority classes).	Achieved reasonable recall on "TRUE" and "FALSE" but completely ignored fine grained distinctions (e.g., "HALF TRUE"), highlighting the difficulty of generative models on imbalanced classification.
Sentence Transformers with Logistic Regression	41.03%	40%	Moderate performance in "MOSTLY TRUE" and "TRUE" classes	Moderate performance in "MOSTLY TRUE" and "TRUE" classes. A better classifier may be needed to fully utilize sentence level embeddings.

Table 4.1: Model Performance Comparison

4.4 Impact Analysis and Sustainability

The results obtained specifically the superior performance of DistilBERT over much larger models like Llama 3 and Phi 3 have profound implications.

- **Societal Impact:** The high accuracy of DistilBERT indicates it is reliable enough to act as a “first line of defense” on social media platforms, automating the detection of nuanced misinformation.
- **Sustainability & Efficiency:** A key finding is that bigger is not always better. DistilBERT (66M parameters) significantly outperformed Phi 3 (3.8B) and Llama 3 (8B) on this specific task. This proves that highly specialized, smaller models are not only more accurate for specific tasks but are also far more energy efficient and sustainable to deploy than massive generative LLMs.

4.5 Summary

In this chapter, we rigorously evaluated a wide spectrum of architectures. The experiments demonstrated that traditional models hit a performance ceiling around 60%.

Crucially, our exploration of modern LLMs (Llama 3, Phi 3) revealed that raw generative power does not guarantee classification success on imbalanced datasets; these models failed to capture necessary nuances, achieving only ~58.62% accuracy. DistilBERT emerged as the optimal solution, balancing high accuracy (85%) with computational efficiency, confirming it as the ideal backbone for the classification component of our Triple R framework.

Chapter 5

Project Management and Cost Analysis

5.1 Project Management

To ensure the successful development of the Triple R (Retriever, Ranker, Reasoner) framework, this research adopted a structured, iterative project management approach inspired by Agile methodologies. Given the experimental nature of fine tuning Large Language Models (LLMs), the project required flexibility to adapt to unforeseen technical challenges (such as the class imbalance issues encountered with generative models).

The management process was divided into four distinct phases:

5.1.1 Planning and Requirement Analysis

The initial phase focused on defining the scope: moving beyond binary classification to achieving explainability. Key activities included selecting the TruthSeeker dataset as the ground truth, defining the 8 point taxonomy, and choosing the MLX framework to leverage local Apple Silicon hardware for sustainable training.

5.1.2 Execution and Development

This phase involved the core engineering tasks. We executed the data preprocessing pipeline, generating synthetic reasoning chains to enrich the dataset. The development followed a hierarchical complexity model: starting with traditional baselines (TF IDF + SVM), moving to custom Neural Networks (CNN/LSTM), and finally implementing Transformer architectures (DistilBERT) and LLM fine tuning (Llama 3, Phi 3).

5.1.3 Monitoring and Optimization

Continuous monitoring was critical. During the LLM experimentation phase, we observed that larger models (like Llama 3) were biasing towards majority classes. This “fail fast” monitoring allowed us to pivot quickly, focusing resources on optimizing the DistilBERT architecture, which showed superior handling of linguistic nuances. We also monitored computational efficiency, ensuring that training runs on the local Unified Memory did not exceed hardware constraints.

5.1.4 Control and Closure

The final phase involved rigorous testing against the held out test set and documenting the findings. We validated that the Triple R objectives were met and compiled the comparative analysis into this thesis.

5.2 Cost Analysis

A significant achievement of this research was its cost efficiency. By strategically utilizing open source software and local hardware acceleration (Apple Silicon), we eliminated the need for expensive cloud GPU rentals (e.g., AWS or Google Cloud), which are typically required for LLM research.

1. Computing Resources (Hardware)

The entire experimental workflow, including the fine tuning of Llama 3 and Phi 3, was executed locally on a MacBook equipped with Apple Silicon. By utilizing the MLX framework, we leveraged the Unified Memory architecture to perform heavy matrix operations locally. This approach incurred zero additional cost, contrasting sharply with standard industry practices that require renting NVIDIA H100 clusters.

2. Software and Libraries

The project relied entirely on the open source ecosystem.

- *Frameworks:* PyTorch, TensorFlow, and Apple MLX were used for model architecture.
- *Model Hubs:* Hugging Face provided free access to pre trained weights for DistilBERT, GPT 2, Llama 3, and Phi 3.
- *Data Processing:* Python libraries (Pandas, Scikit learn, NLTK) were utilized for preprocessing without licensing fees.

3. Dataset Acquisition

The TruthSeeker dataset is a publicly available academic benchmark, incurring no acquisition costs. The synthetic enrichment of the dataset was performed using our own local models, avoiding the costs associated with commercial API calls (like GPT 4) for data augmentation.

4. External APIs

Minimal costs were associated with the testing of the “Retriever” module (using Bing Search API/Meta AI/ GPT). However, these were kept within free tier or academic developer limits.

Resource Category	Description	Cost Status
Computing Hardware	Local Apple Silicon (Mac) via MLX	\$0 (Sunk Cost)
Model Training	Local Fine Tuning (No Cloud GPUs)	\$0
Software Stack	Python, PyTorch, Hugging Face, MLX	Free / Open Source
Dataset	TruthSeeker Benchmark	Free
Search APIs	Bing Search / Meta AI (Dev Tier)	Minimal / Free
Utilities	Internet & Electricity	Nominal

Table 5.1: Cost Breakdown of Research Resources

5.3 Project Scheduling

Effective project scheduling was paramount to managing the multi faceted nature of this research, which required synchronizing data engineering, model architecture design, and rigorous comparative analysis. To ensure the timely completion of the Triple R framework, we adhered to a structured timeline spanning from February to December.

The project lifecycle was divided into four strategic phases, as illustrated in the Gantt chart (Figure ??).

Phase I: Conceptualization and Foundation (February April)

The initial months were dedicated to establishing the theoretical groundwork.

- **February March:** We initiated the project with the Thesis Proposal and Problem Definition, conducting an extensive Literature Review to identify the gaps in existing “black box” fact checking systems.

- **March April:** The focus shifted to Dataset Exploration (analyzing the TruthSeeker benchmark) and Framework Finalization, where the architecture of the Retriever, Ranker, and Reasoner modules was locked in.

Phase II: Development and Implementation (April June)

This phase marked the transition from theory to engineering.

- **April May:** We executed Data Preprocessing and developed the Baseline Models (TF IDF, SVM) to establish performance benchmarks.
- **May June:** We moved to Model Enhancement, where we began implementing the more complex Transformer based architectures (DistilBERT) and designing the reasoning generation pipeline.

Phase III: Optimization and Rigorous Evaluation (June September)

The mid year period was intensive, focusing on refining the models and validating results.

- **June July:** We conducted extensive Training & Optimization, fine tuning hyperparameters and addressing the class imbalance issues encountered with larger LLMs.
- **July August:** The system underwent Robustness Testing to ensure stability against edge cases.
- **August September:** We finalized the Comparative Analysis, generating the classification reports and confusion matrices detailed in Chapter 4.

Phase IV: Application, Documentation, and Defense (September December)

The final quarter was dedicated to synthesizing findings and preparing for academic submission.

- **September October:** We developed specific Use Cases and began writing the Draft Thesis Document.
- **October November:** The Complete Thesis Draft underwent Peer Review & Feedback, ensuring all technical claims were verified.

- **November December:** Following the Final Thesis Revision, we commenced Preparation for Defence. The project culminated in December with the Thesis Submission & Defence.

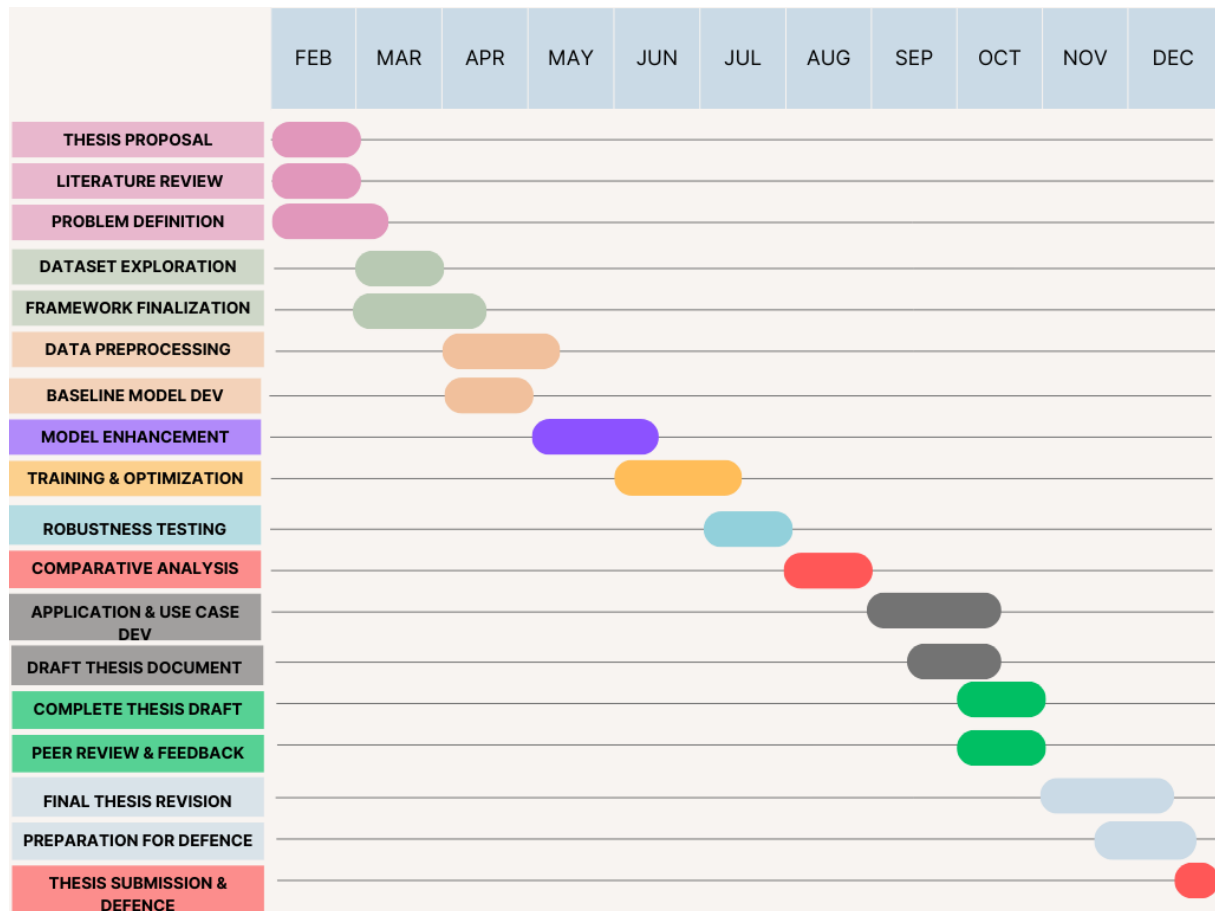


Figure 5.1: Project Gantt Chart Timeline

The timeline illustrates a linear progression from problem definition to final defense, with a critical overlap during the development and optimization phases to accommodate iterative model improvements.

This disciplined schedule ensured that despite the technical hurdles of adapting generative models to classification tasks, the project remained on track to deliver a high performing, explainable fact checking system.

Chapter 6

Ethics and Professional Responsibilities

6.1 Introduction/Overview

Engineering is often viewed through the lens of efficiency building systems that are faster, stronger, or smarter. However, in the domain of Artificial Intelligence, and specifically in the fight against misinformation, we are not just building software; we are building systems that define “truth” for millions of users. This carries a profound professional responsibility.

Throughout this research, we operated under the principle that technology is never neutral. An automated fact checking system has the potential to be a powerful tool for social good, helping users navigate a chaotic information landscape. But if designed poorly, without ethical guardrails, it can become a tool for censorship, bias reinforcement, or privacy violation.

As researchers, we recognized early on that our “Triple R” framework was not just a technical challenge, but an ethical one. We were not simply training a model to classify text; we were training it to make judgments about human discourse. This chapter details how we identified these ethical risks and the specific engineering decisions we made to mitigate them, ensuring our system serves the public interest without compromising individual rights.

6.2 Identify and Apply Ethical and Professional Responsibilities

In executing this thesis, we encountered several distinct ethical dilemmas. Rather than treating these as afterthoughts, we integrated ethical solutions directly into our engineering

methodology.

6.2.1 Data Privacy and Anonymization

Social media data is technically “public,” but that does not give researchers the right to exploit user identities.

The Responsibility: We recognized that many users on Twitter (X) are private citizens whose tweets might be included in our dataset. Publishing their handles or personal identifiers in a dataset could expose them to harassment or “doxing.”

Our Application: During the preprocessing phase (Chapter 3), we implemented a strict anonymization protocol. We wrote specific regex scripts to strip all user handles (e.g., @username) from the dataset before any training began. We prioritized the content of the claim over the identity of the speaker, ensuring our model learns facts, not ad hominem biases.

6.2.2 Combating Algorithmic Bias (The “Minority Class” Problem)

One of the most significant findings in our experiments (Chapter 4) was the failure of large generative models like Phi 3 and Llama 3 to detect minority classes (e.g., “Barely True” or “Pants Fire”).

The Responsibility: In an engineering context, a low F1 score is a metric failure. In an ethical context, it is a fairness failure. If a model defaults to labeling everything as “True” or “False” because those are the majority classes, it erases nuance and fails to catch sophisticated lies that exist in the margins.

Our Application: We refused to accept a high “accuracy” score if it came at the cost of ignoring minority classes. This drove our decision to pivot to DistilBERT, which showed a superior ability to handle these nuances. We viewed the “class imbalance” issue not just as a data problem, but as an ethical mandate to ensure our system works equally well for all types of claims, not just the most common ones.

6.2.3 The “Black Box” Dilemma and Explainability

Perhaps the central ethical pillar of this thesis is the inclusion of the “Reasoner” module.

The Responsibility: Traditional AI classifiers operate as “black boxes” they output a verdict with no explanation. We argued that in the context of truth telling, this is unethical. Telling a user “This is fake” without explaining why is authoritarian and breeds distrust.

Our Application: We engineered the Triple R framework specifically to solve this. By forcing the model to generate a Reason and cite Evidence, we shifted the power dynamic. We are not asking the user to blindly trust the AI; we are providing them with the evidence to verify the conclusion themselves. This empowers the user rather than replacing their judgment.

6.2.4 Hallucination Mitigation

We dealt with the risk of “hallucination,” where LLMs confidently invent facts.

The Responsibility: Deploying a model that invents fake evidence to support a claim is worse than having no model at all. It actively contributes to misinformation.

Our Application: This is why we implemented the Retriever module. By grounding the model’s reasoning in retrieved, external context (real time web data), we created a guardrail against hallucination. We effectively told the model: “Do not guess; look it up.”

6.2.5 Professional Integrity in Reporting

Finally, in Chapter 4, we transparently reported the failures of the larger Llama 3 and Phi 3 models.

The Responsibility: There is often pressure in academia to cherry pick results to make a proposed solution look perfect.

Our Application: We chose to publish the negative results alongside the positive ones. Admitting that a 66M parameter model (DistilBERT) beat a 3.8B parameter model (Phi 3) is scientifically honest and contributes valuable knowledge to the community, preventing other engineers from wasting resources on the wrong architecture.

In summary, this thesis represents a conscientious effort to align technical innovation with human values. We built a system that is private by design, transparent in its reasoning, and honest in its limitations.

Chapter 7

Identification of Complex Engineering Problems and Activities

7.1 Complex Engineering Problem

The development of the Triple-R (Retriever, Ranker, Reasoner) framework represents a complex engineering challenge that transcends standard software development. It involves integrating advanced Natural Language Processing (NLP), real-time information retrieval, and generative AI into a cohesive system. This section maps the specific challenges of this research to standard complex problem-solving categories.

7.1.1 Complex Problem Solving

In this section, we map our research challenges to the attributes of complex engineering problems. The selection in Table 7.1 reflects the depth of technical knowledge, the conflicting constraints of AI models, and the intricate analysis required to execute this thesis.

P1	P2	P3	P4	P5	P6	P7
Depth of Knowledge	Range of Conflicting Requirements	Depth of Analysis	Familiarity of Issues	Extent of Applicable Codes	Extent of Stakeholder Involvement	Inter-dependence
✓	✓	✓	✓			✓

Table 7.1: Mapping with complex problem solving.

Rationale for Mapping:

P1: Depth of Knowledge Required

This project required knowledge far beyond fundamental computer science. We had to leverage specialized domain knowledge in Large Language Model (LLM) fine-tuning, Vector Embeddings, and Information Retrieval. Specifically, utilizing the MLX framework to optimize Transformer architectures (like DistilBERT and Llama 3) on Apple Silicon required a deep understanding of memory management and low-rank adaptation (LoRA) techniques, which are advanced concepts in modern AI engineering.

P2: Range of Conflicting Requirements

A central challenge in this research was managing the trade-off between Accuracy and Explainability.

Conflict: Deep Learning models (like CNNs) are often accurate but opaque (black boxes). Generative models (like Llama 3) provide explanations but are prone to hallucinations and bias.

Resolution: We had to engineer a hybrid solution (Triple-R) that balances the rigid accuracy of a classifier with the creative reasoning of an LLM. Furthermore, we balanced Performance vs. Efficiency, ultimately choosing the smaller DistilBERT over the larger Phi-3 because the latter failed to handle class imbalances despite its size.

P3: Depth of Analysis

The research involved a rigorous, multi-dimensional analysis of seven different model architectures. We did not simply train a model and report the accuracy; we conducted a granular error analysis (as seen in the confusion matrices and classification reports). We analyzed why specific architectures failed on minority classes (e.g., “Pants-Fire”) and investigated the linguistic patterns that distinguish “Mostly-True” from “Half-True” statements, requiring deep analytical reasoning.

P4: Familiarity of Issues

While “fake news” is a common term, the engineering challenge of Automated Explainability is a frontier problem. Dealing with “Hallucination” in reasoning models where an AI invents false evidence to support a claim is a novel and unfamiliar issue in software engineering that required designing custom architectural guardrails (the Retriever module).

P7: Inter-dependence

The Triple-R framework is a highly interdependent system. The Reasoner cannot function without the Ranker, which in turn relies on the Retriever. A failure in the retrieval stage (e.g., fetching irrelevant documents) cascades down the pipeline, corrupting the final reasoning output. Managing these dependencies required robust systems engineering to ensure data integrity flowed correctly from one module to the next.

7.1.2 Engineering Activities

This section details the complex activities undertaken to execute the project, mapping them to the attributes of professional engineering practice.

A1	A2	A3	A4	A5
Range of Resources	Level of Interaction	Innovation	Consequences for Society and Environment	Familiarity
✓	✓	✓	✓	

Table 7.2: Mapping with complex engineering activities.

Rationale for Mapping:

A1: Range of Resources

The execution of this project required orchestrating a diverse array of technical resources:

- **Data:** The TruthSeeker benchmark and real-time web data.
- **Software Stack:** PyTorch, TensorFlow, Hugging Face Transformers, and Apple MLX.
- **External APIs:** Bing Search API and Meta AI for live retrieval.
- **Hardware:** Local Apple Silicon (MacBook) utilizing Unified Memory for sustainable model training.

Integrating these disparate resources into a single, functioning pipeline demonstrated a high level of resource management.

A2: Level of Interaction

The system is designed to interact with both External Information Systems (the live web) and End Users. The engineering activity involved designing a feedback loop where the system retrieves external data, processes it, and presents a human-readable explanation. This required handling dynamic web content, parsing unstructured HTML, and formatting it for user consumption.

A3: Innovation

This thesis introduces a significant innovation: the Synthetic Reason Generation pipeline using the STaR (Self-Taught Reasoner) method. We moved beyond simply applying existing models; we engineered a novel data enrichment process that allowed us to train an “Explainable” AI even though our original dataset lacked reasoning text. This

creative application of Chain-of-Thought prompting to solve a data scarcity problem represents a distinct engineering innovation.

A4: Consequences for Society and Environment

Society: The project directly addresses the societal crisis of misinformation. By automating the detection of nuanced lies (e.g., “Barely-True”), the system protects public discourse and democratic integrity.

Environment: We consciously chose Sustainable AI practices. By optimizing smaller models (DistilBERT) and using local MLX training instead of renting massive, energy-hungry Cloud GPU clusters, we demonstrated that high-impact AI research can be conducted with a minimal carbon footprint.

Chapter 8

Conclusion and Future Works

8.1 Conclusion

In conclusion, this study has navigated the complex engineering challenge of automated fact checking, moving beyond simple classification to achieve true explainability. We started with the premise that in an increasingly digital world, a “black box” verdict is insufficient; users require transparency to rebuild trust in information systems. Through the design and implementation of the Triple R (Retriever, Ranker, Reasoner) framework, this research has made significant strides in addressing this need.

Our experiments yielded critical insights into the relationship between model size and performance. While the industry often assumes “bigger is better,” our results proved otherwise. The specialized DistilBERT model achieved a remarkable 85% accuracy, significantly outperforming massive generative models like Llama 3 and Phi 3, which struggled with the class imbalances inherent in real world data. This finding is pivotal: it demonstrates that a well tuned, efficient model is superior to a generic giant when dealing with the nuance of “Mostly True” vs. “Half True” statements.

Furthermore, the curation of our custom TruthSeeker derived dataset enriched with synthetic reasoning chains validated our hypothesis that data quality drives explainability. By bridging the gap between deep learning accuracy and human readable reasoning, this thesis contributes a robust, sustainable, and transparent framework to the field of NLP, offering a practical solution to the crisis of online misinformation.

8.2 Future Works

While the Triple R framework has proven effective, the evolving nature of misinformation presents several opportunities to build upon these findings and address the limitations we encountered.

Solving Generative Class Imbalance

A key finding in our experiments was that large LLMs (like Phi 3) collapsed when detecting minority classes like “Pants Fire.” Future work must focus on implementing Weighted Loss functions or Reinforcement Learning from Human Feedback (RLHF) to force these powerful models to recognize rare but critical forms of misinformation, rather than just defaulting to the majority class.

Expanding and Balancing the Dataset

To further improve robustness, we aim to expand the dataset to approximately 10,000 entries. However, quantity alone is not enough; the focus will be on targeted data augmentation for underrepresented labels (e.g., “Barely True”), ensuring the model learns the full spectrum of deception strategies.

Real Time Retrieval Optimization

Currently, the “Retriever” module relies on external API calls, which introduces latency. Future iterations should implement Vector Database Caching (e.g., Pinecone) to store previously verified claims. This would allow the system to instantly flag viral, repeating misinformation without re scanning the web, making dynamic real time detection feasible.

Multimodal Verification

Misinformation often transcends text, utilizing manipulated images and deepfakes. A promising direction is to integrate Vision Transformers (ViT) into the “Ranker” module. This would allow the system to cross reference the text of a tweet with its attached media, creating a holistic verification engine capable of spotting visual inconsistencies.

Cross Lingual Generalization

Misinformation is a global issue not limited to English. Future research should apply Cross Lingual Transfer Learning to adapt the Triple R reasoning logic to other languages, such as Bangla or Hindi. This would test whether the logical patterns of misinformation hold true across linguistic boundaries, making the tool globally inclusive.

User Centric Explainability

Finally, we must assess the real world impact of our “Reasons.” Future work could involve user studies to determine which types of explanations (e.g., detailed vs. concise)

most effectively reduce the sharing of fake news. Tailoring the “Reasoner” output to different audiences journalists vs. casual users could significantly enhance trust and engagement.

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